

Information Technology Project Management

Chapter 1

INTRODUCTION TO PROJECT MANAGEMENT

Note: See the text itself for full citations (the 7th-9th editions).
Some content has been updated from the 9th edition.

Learning Objectives

- ▶ Understand the growing need for better project management, especially for IT projects.
- ▶ Explain what a project is, list various attributes of projects, and describe the triple constraint of project management.
- ▶ Describe project management and discuss key elements of the project management framework, including project stakeholders, the project management knowledge areas, common tools and techniques, and project success.

Learning Objectives

- ▶ Discuss the relationship between project and program, and the contributions each makes to enterprise success.
- ▶ Understand the role of project managers by describing what they do, and what skills they need.
- ▶ Describe the project management profession, including its history, the role of professional organizations like the Project Management Institute (PMI), and the importance of certification and ethics.

Introduction

- ▶ Many people and organizations today have a new or renewed interest in project management.
- ▶ Computer hardware, software, networks, and the use of interdisciplinary and global work teams have radically changed the work environment.
- ▶ Worldwide IT spending was \$3.5 trillion in 2017, a 2.4 percent increase from 2016 spending.
- ▶ The Project Management Institute reported that the number of project-related jobs reached almost 66 million in 2017, and demand continues to increase.
“By 2027, employers will need 87.7 million individuals working in project management–oriented roles.”

Project Management Statistics

- ▶ The unemployment rate for IT professionals is generally half the rate of the overall labor market in the US.
- ▶ In 2017, the average annual salary (without bonuses) for someone in the project management profession was \$130.866 in Switzerland, the highest-paid country.
 - In 2025, IT project managers can expect to earn from 30 million to 70 million VND monthly in Vietnam.
 - Salaries of survey respondents across 37 counties were 23 percent higher for those with the Project Management Professional (PMP) credential than for those without it.
- ▶ The top skills employers look for in new college graduates are all related to project management: teamwork, decision-making, problem-solving, and verbal communications.

Advantages of Using Project Management Techniques

- ▶ Better control of financial, physical, and human resources
- ▶ Improved customer relations
- ▶ Shorter development times
- ▶ Lower costs and improved productivity
- ▶ Higher quality and increased reliability
- ▶ Higher profit margins
- ▶ Better internal coordination
- ▶ Positive impact on meeting strategic goals
- ▶ Higher worker morale

What Is a Project?

- ▶ A **project** is “a temporary endeavor undertaken to create a unique product, service, or result.” (*PMBOK® Guide*)
- ▶ Projects end when their objectives have been reached or the project has been terminated.
- ▶ Projects can be large or small and take a short or long time to complete.

Examples of IT Projects

- ▶ A large network of healthcare providers updates its information systems and procedures to reduce hospital-acquired diseases.
- ▶ A team of students creates a smartphone application and sells it online.
- ▶ A company develops a driverless car.
- ▶ A company implements a new system to increase sales force productivity and customer relationship management that will work on various laptops, smartphones, and tablets.
- ▶ A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus.
- ▶ A government group develops a system to track child immunizations.

Project Attributes

- ▶ A project:
 - *has a unique purpose,*
 - *is temporary,*
 - *drives change and enables value creation,*
 - *is developed using progressive elaboration,*
 - *requires resources, often from various areas,*
 - *should have a primary customer or sponsor,*
 - The **project sponsor** usually provides the direction and funding for the project.
 - *involves uncertainty.*

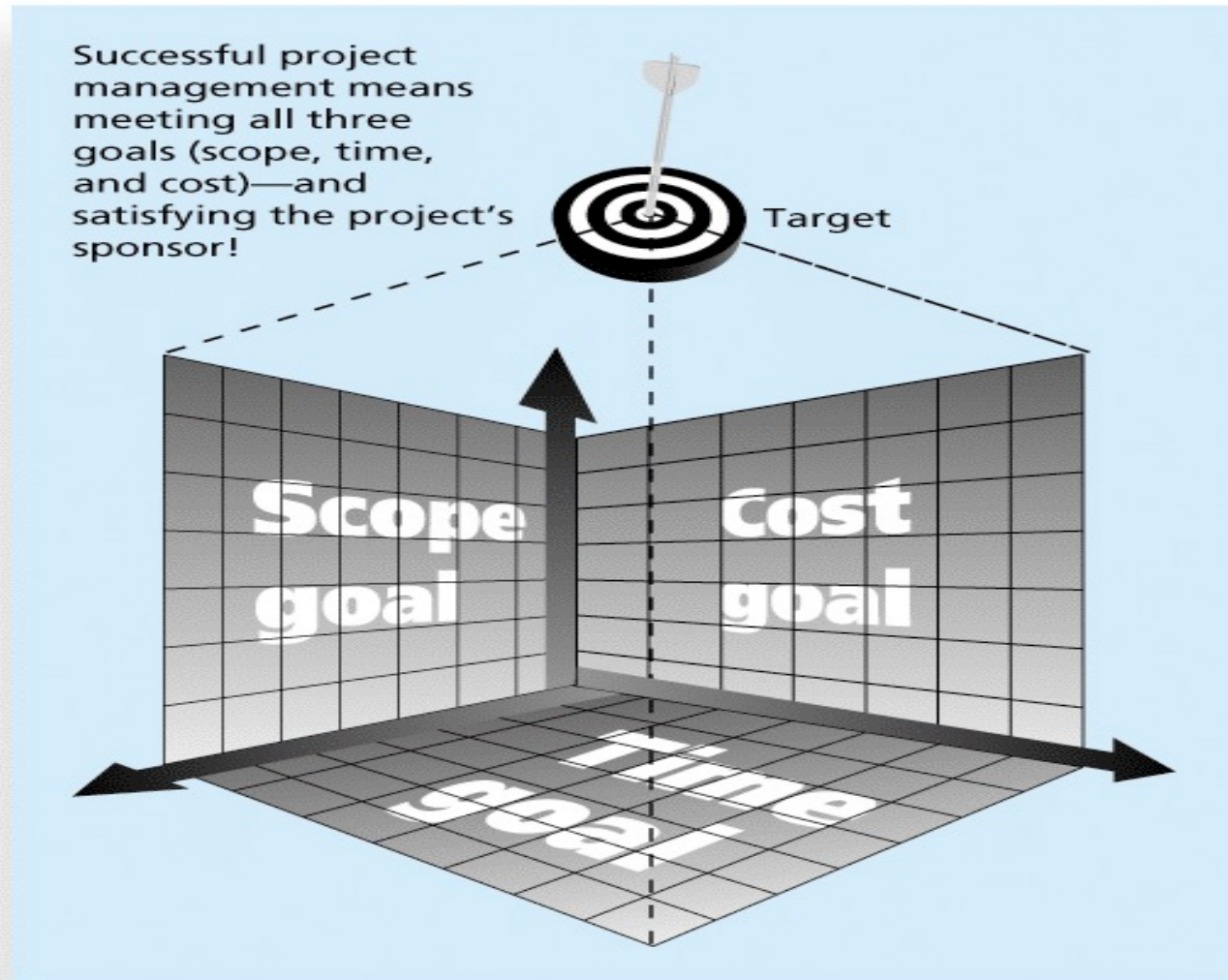
Project and Program Managers

- ▶ **Project managers** work with project sponsors, the project team, and other people involved in a project to meet project goals.
- ▶ **Program:** A group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually. (*PMBOK® Guide*)
- ▶ Program managers oversee programs; they often act as bosses for project managers.

Project Constraints

- ▶ To create a successful project, a project manager must consider scope, time, cost and balance these three goals.
 - **Scope:** What work will be done as part of the project? What unique product, service, or result does the customer or sponsor expect from the project? How will the scope be verified?
 - **Time:** How long should it take to complete the project? What is the project's schedule? How will the team track actual schedule performance? Who can approve changes to the schedule?
 - **Cost:** What should it cost to complete the project? What is the project's budget? How will costs be tracked? Who can authorize changes to the budget?

FIGURE 1-1 Project Constraints



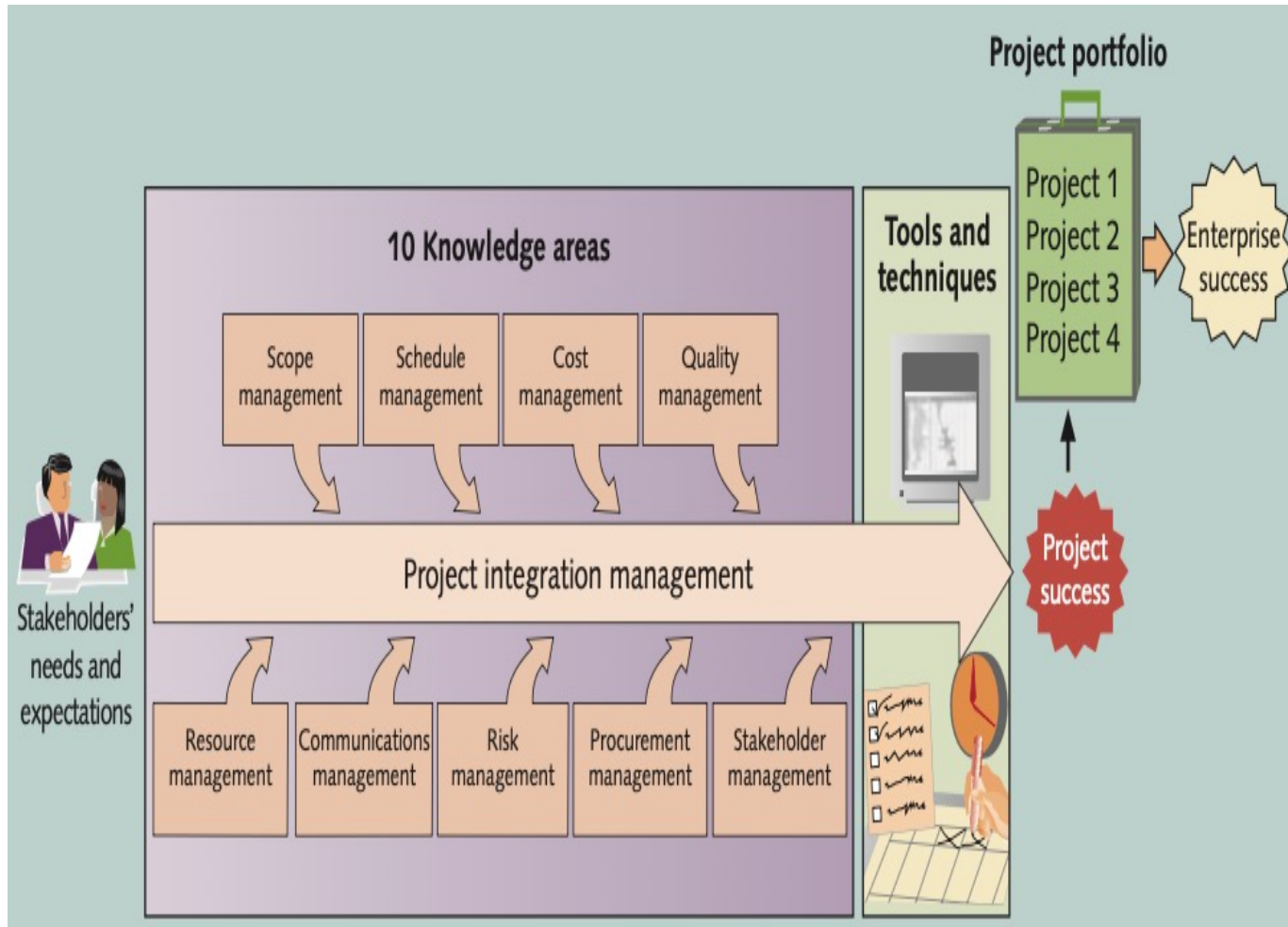
What Is Project Management?

- ▶ **Project management** is “the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.” (*PMBOK® Guide*)
- ▶ Project managers must not only strive to meet specific scope, time, cost, and quality goals of projects, but also facilitate the entire process to meet the needs and expectations of project stakeholders.

Project Stakeholders

- ▶ **Stakeholders** are the people involved in or affected by project activities.
- ▶ Stakeholders include the project sponsor, the project team, support staff, customers, users, suppliers, and even opponents of the project.

FIGURE 1-2 Project Management Framework



Project Management Knowledge Areas

- ▶ **Project management knowledge areas** describe the key competencies that project managers must develop.
 - Project scope management involves defining and managing all the work required to complete the project successfully.
 - Project schedule management (formerly called project time management) includes estimating how long it will take to complete the work, developing an acceptable project schedule, and ensuring the timely completion of the project.
 - Project cost management consists of preparing and managing the budget for the project.
 - Project quality management ensures that the project will satisfy the stated or implied needs for which it was undertaken.

Project Management Knowledge Areas

- ▶ **Project management knowledge areas** describe the key competencies that project managers must develop.
 - Project resource management is concerned with making effective use of the people and physical resources involved with the project.
 - Project communications management involves generating, collecting, disseminating, and storing project information.
 - Project risk management includes identifying, analyzing, and responding to risks related to the project.
 - Project procurement management involves acquiring or procuring goods and services for a project from outside the performing organization.

Project Management Knowledge Areas

- ▶ **Project management knowledge areas** describe the key competencies that project managers must develop.
 - Project stakeholder management involves identifying and analyzing stakeholder needs, while also managing and controlling their engagement throughout the project's life cycle.
 - Project integration management is an overarching function that affects and is affected by all of the other knowledge areas.
- ▶ Project managers must have knowledge and skills in these 10 knowledge areas.

Project Management Tools and Techniques

- ▶ **Project management tools and techniques** assist project managers and their teams in various aspects of project management.
- ▶ Some specific ones include:
 - Statements of work, scope management plans, scope verification techniques, and scope change controls (**scope management**).
 - Project network diagrams, critical path analysis, crashing, fast tracking, and schedule performance measurements (**schedule management**).
 - Project budgets, net present value, return on investment, payback analysis, earned value management, cost estimates, cost management plans, and cost baselines (**cost management**).

Super Tools

- ▶ “**Super tools**” are those tools that have high use and high potential for improving project success, such as:
 - Project management software, change requests, and lessons-learned reports (**integration management**).
 - Scope statements, work breakdown structures, and requirements analyses (**scope management**).
 - Gantt charts (**schedule management**).

Other Tools

- ▶ *PMBOK® Guide* (sixth edition) lists tools and techniques based on their purpose:
 - **Data gathering:** benchmarking, brainstorming, check sheets, checklists, focus groups, interviews, market research, questionnaires and surveys, and statistical sampling.
 - **Data analysis:** alternatives analysis, assessment of other risk parameters, assumption and constraint analysis, cost of quality, cost-benefit analysis, decision tree analysis, document analysis, and earned value analysis.
 - **Data representation:** affinity diagrams, cause-and-effect diagrams, control charts, flow charts, hierarchical charts, histograms, logical data models, matrix diagrams, matrix-based charts, mind mapping, probability and impact matrix, scatter diagrams, stakeholder engagement assessment matrix, stakeholder mapping/representation, and text-oriented formats.

Other Tools

- ▶ *PMBOK® Guide* (sixth edition) lists tools and techniques based on their purpose:
 - **Decision making:** multi-criteria decision analysis and voting.
 - **Communication:** feedback and presentations.
 - **Interpersonal and team skills:** active listening, communication styles assessment, conflict management, cultural awareness, decision making, emotional intelligence, facilitation, influencing, leadership, meeting management, motivation, negotiation, networking, nominal group, observation/conversation, political awareness, and team building.

Project Success

- ▶ There are several ways to define project success:
 - *The project met scope, time, and cost goals.*
 - *The project satisfied the customer/sponsor.*
 - *The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy.*

What Helps Projects Succeed?

- Executive sponsorship
- Emotional maturity
- User involvement
- Optimization
- Skilled resources
- Agile process
- Modest execution
- Project management expertise
- Clear business objectives

Source: The Standish Group, “CHAOS Manifesto 2015” (2015).

Program Managers

- ▶ A **program manager** provides leadership and direction for the project managers heading the projects within the program.
- ▶ Examples of common programs in the IT field include infrastructure, application development, and user support.

The Role of the Project Manager

The job description for a project manager can vary by industry and by organization, but most project managers perform similar tasks (planning, scheduling, coordinating, and working with people to achieve project goals).

Suggested Skills for Project Managers

- ▶ Project managers need to have a wide variety of skills and be able to decide which skills are more important in different situations.
- ▶ A project manager should:
 - possess project management knowledge,
 - understand the project (project environment knowledge, application area knowledge, standards, and regulations) and how it will fit into the organization,
 - experience in general management and human relations, or soft skills needed to guide the project team.

Ten Hot Tech Skills for 2017

1. Full-stack software development
2. Project management
3. Cyber-security
4. Networking
5. User experience/user interface (UX/UI) design
6. Quality assurance (QA)/testing
7. Cloud engineering
8. Big data
9. Machine learning/artificial intelligence
10. DevOps

Source: Sharon Florentine, “10 IT skills that employers need in 2017,”
CIO from IDG (February 1, 2017).

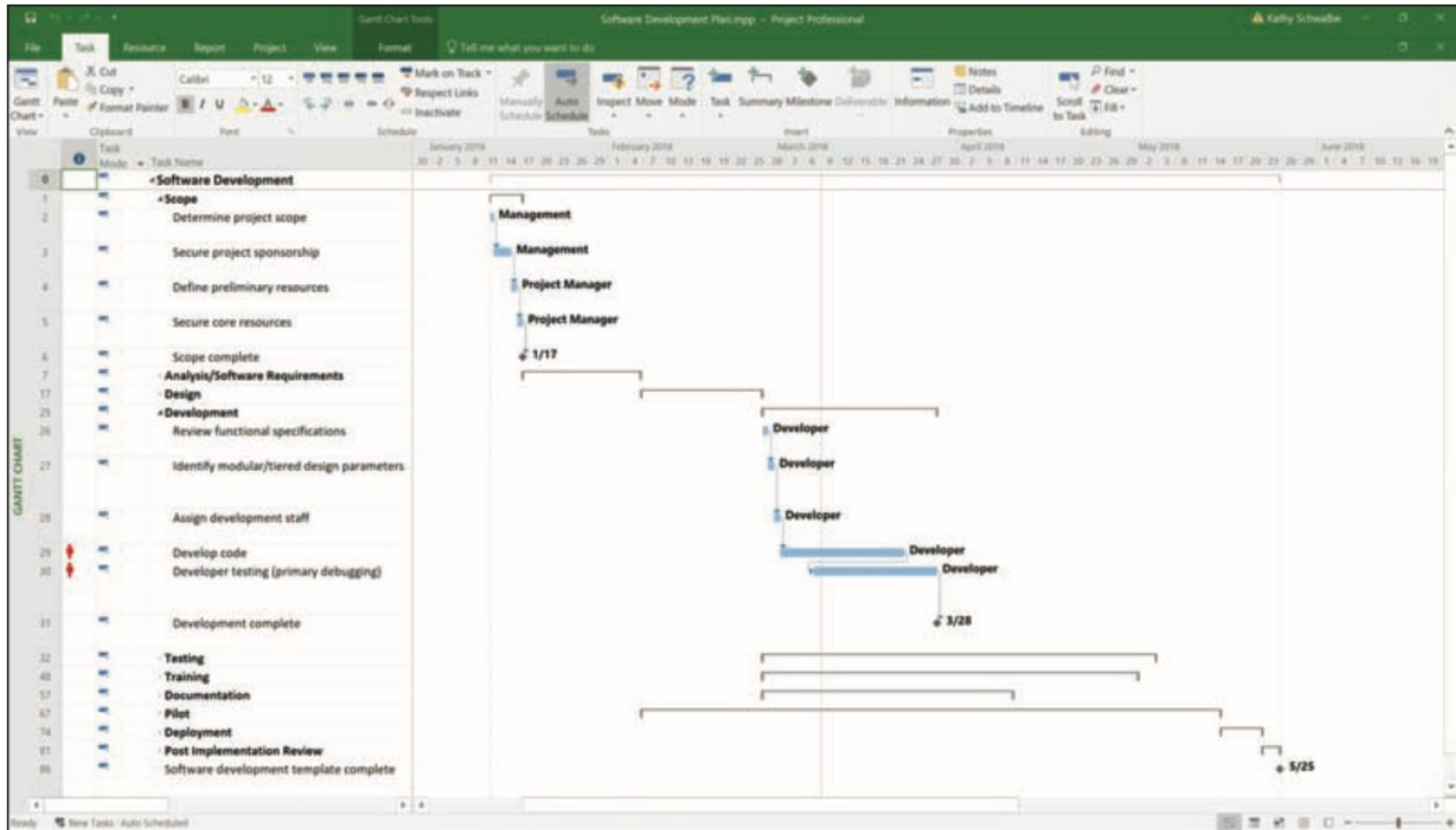
Importance of Leadership Skills

- ▶ A **leader** focuses on long-term goals and big-picture objectives, while inspiring people to reach those goals.
- ▶ A **manager** often deals with the day-to-day details of meeting specific goals.
- ▶ Project managers often take on the role of both leader and manager.

The Project Management Profession

- ▶ Most people agree that the modern concept of project management began with the Manhattan Project, which the U.S. military led to develop the atomic bomb during World War II:
 - The project involved many people with different skills at several different locations.
 - The project clearly separated the overall management of the project's mission, schedule, budget, and techniques.
 - It lasted about three years and cost almost \$2 billion in 1946.

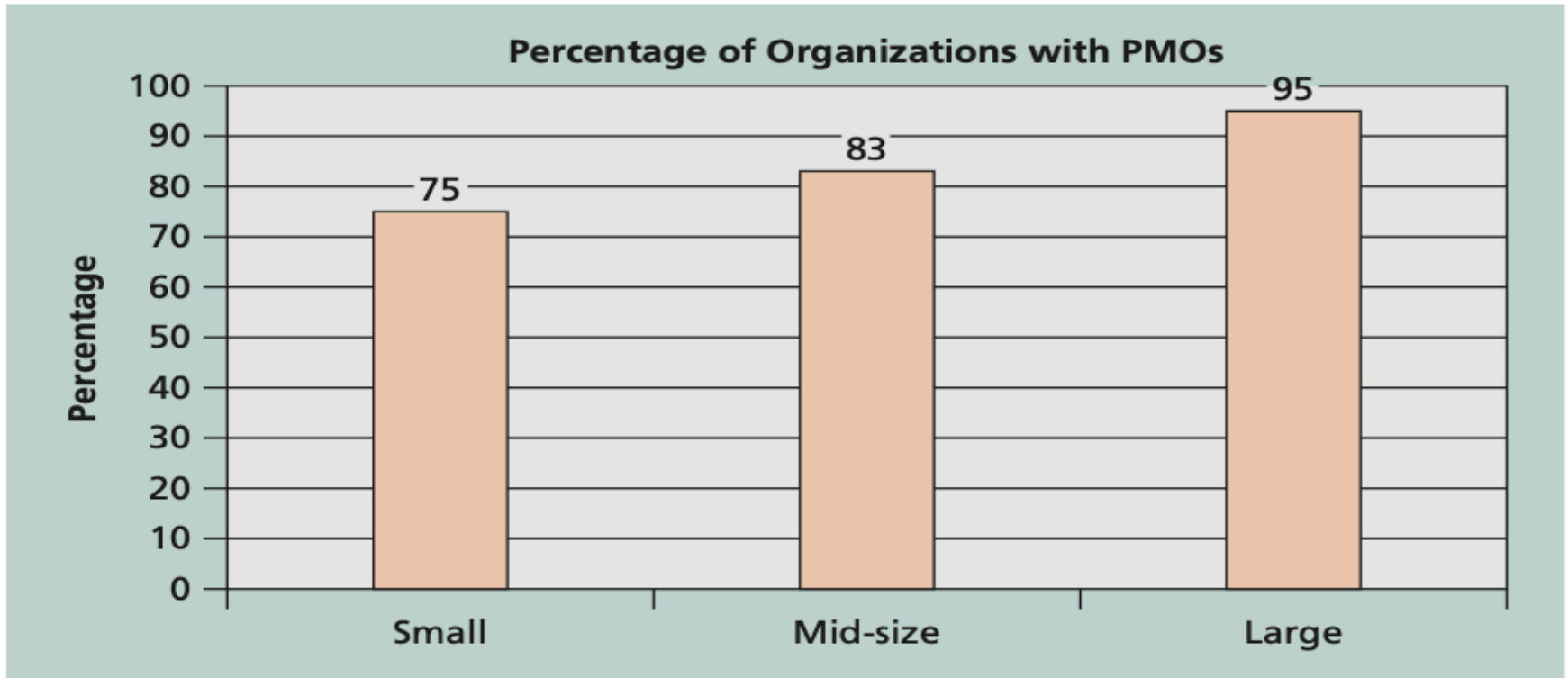
Sample Gantt Chart Created with Project 2016



Project Management Offices

- ▶ In the 1990s, many companies began creating PMOs to help them handle the increasing number and complexity of projects.
- ▶ A **Project Management Office (PMO)** is an organizational group responsible for coordinating the project management function throughout an organization.

Percentage of Organizations with PMOs

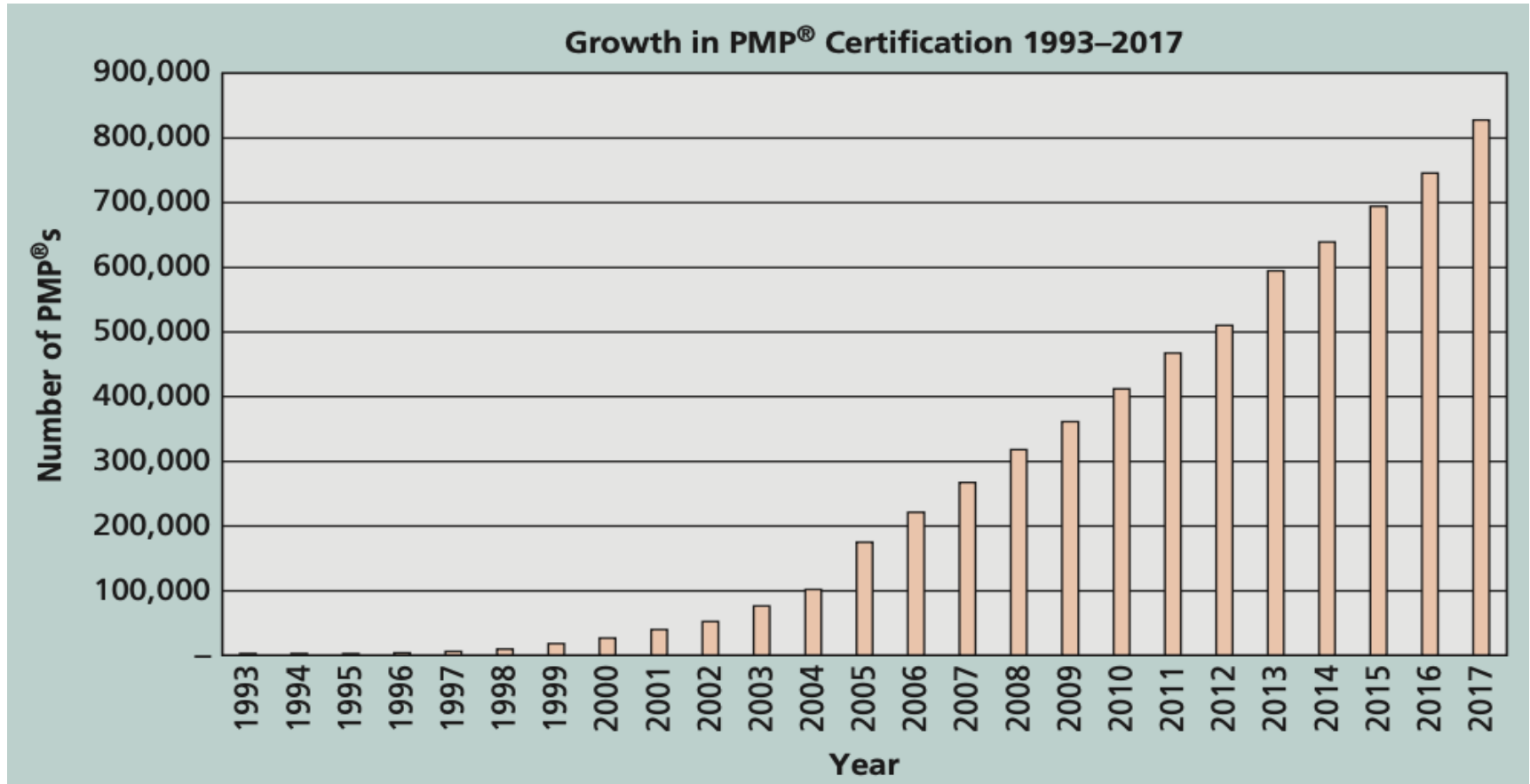


Source: PM Solutions, "The State of the Project Management Office (PMO) 2016," 2016.

The Project Management Institute

- ▶ The Project Management Institute (PMI) is an international professional society for project managers, founded in 1969.
 - PMI has continued to attract and retain members, reporting more than 500,000 members worldwide by the end of 2017.
 - PMI provides certification as a **Project Management Professional (PMP)**—someone who has documented sufficient project experience and education, agreed to follow the PMI code of professional conduct, and demonstrated knowledge of project management by passing a comprehensive examination.

Growth in PMP Certification 1993-2017



Source: Data points from Project Management Institute, PMI Today (annual issues).

